Purpose of this experiment

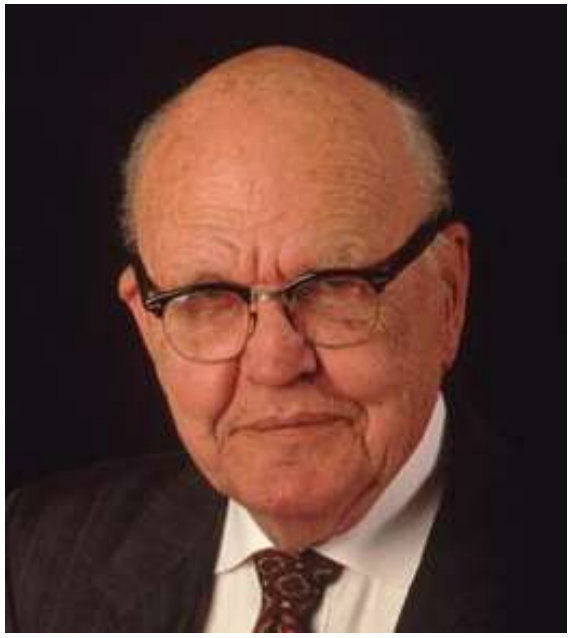
Learn briefly about MOSFET which is used in integrated circuit (IC).

Experience one end of fabrication process.

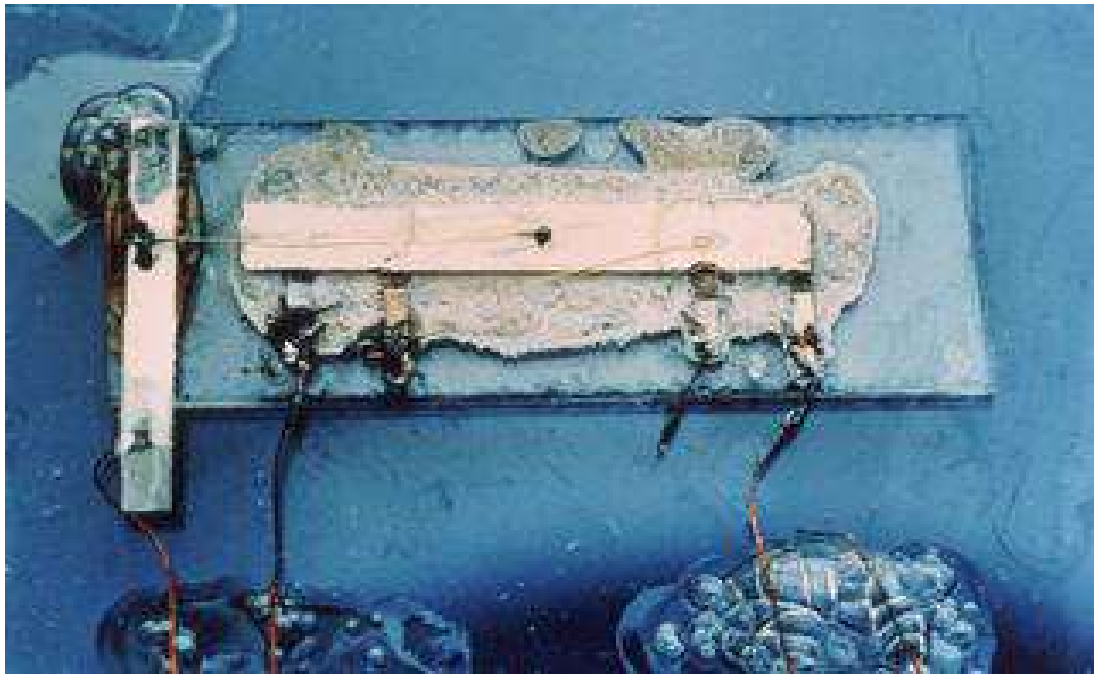
Experience electrical analysis on MOS capacitor which is a main part of MOSFET.

What is integrated circuit (IC)？

A network of circuit elements on a substrate (conventionally Si wafer).



Jack Kilby, 2000 Novel Prize winner.



The first IC invented by J. Kilby (Texas Instruments) in 1958.

Si wafer

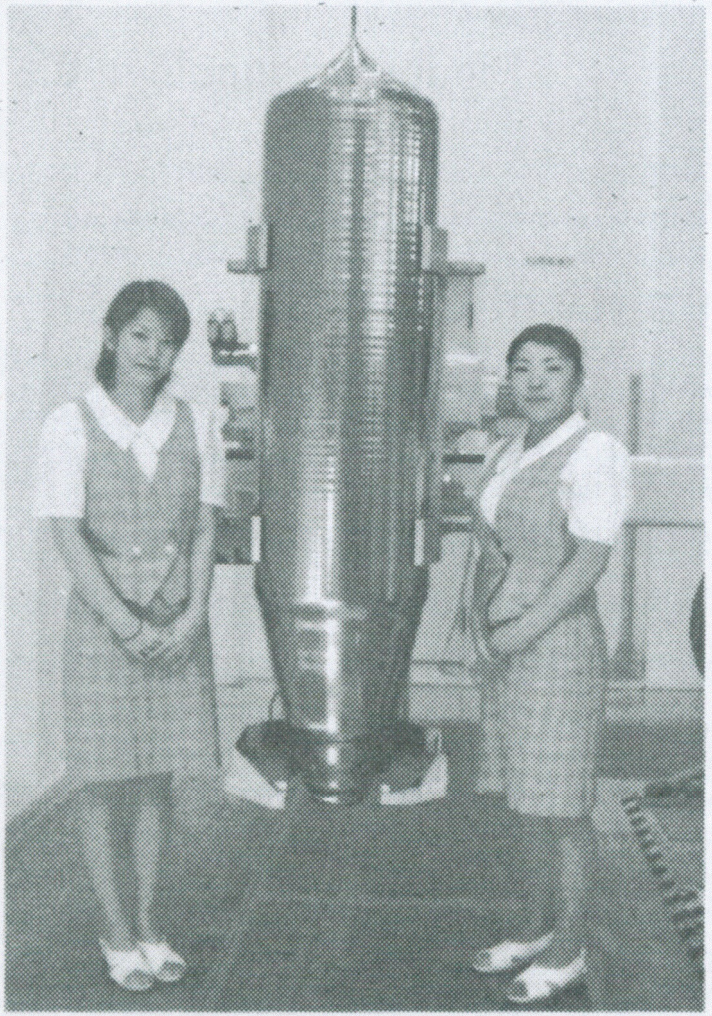
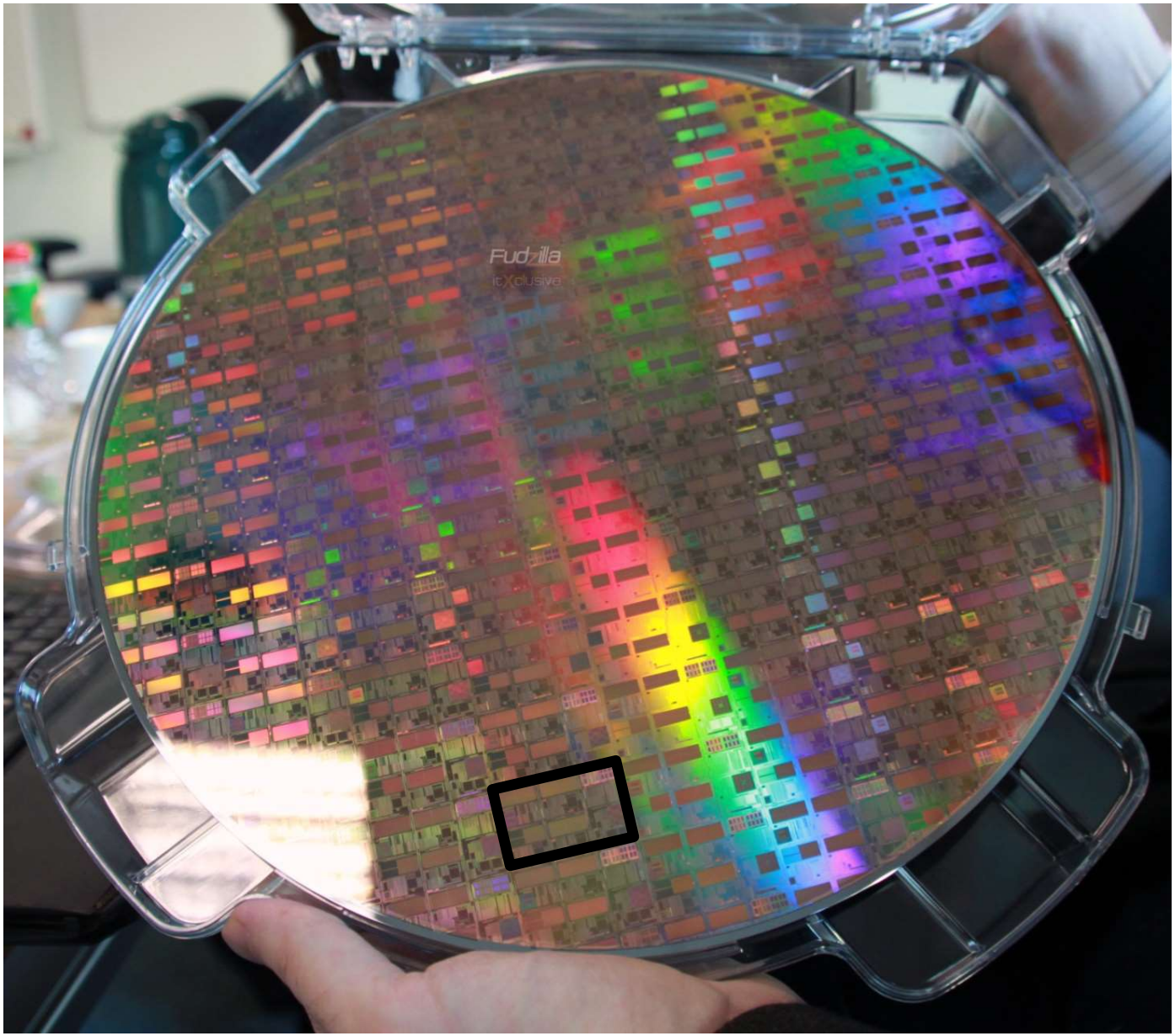
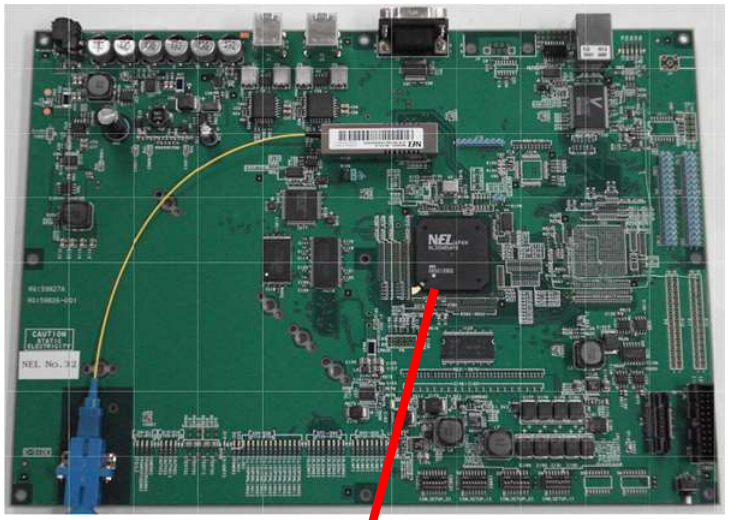


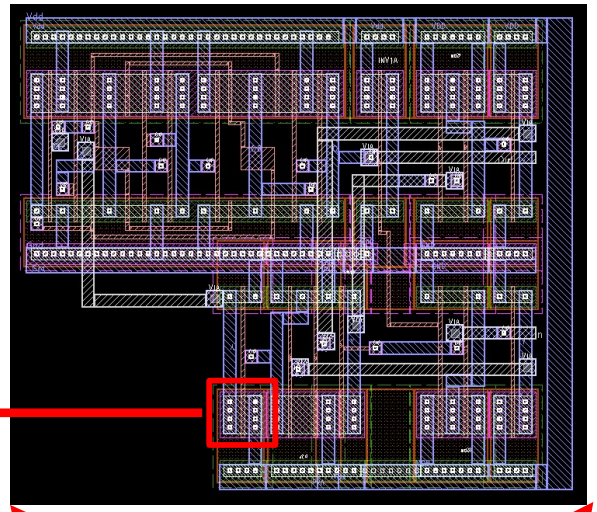
Figure 5: The heaviest DF crystal:
body length = 1100 mm, weight = 438 kg

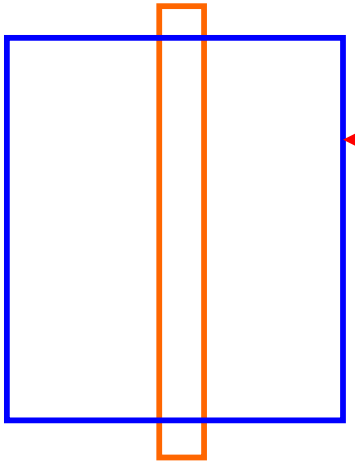
IC on a Si wafer

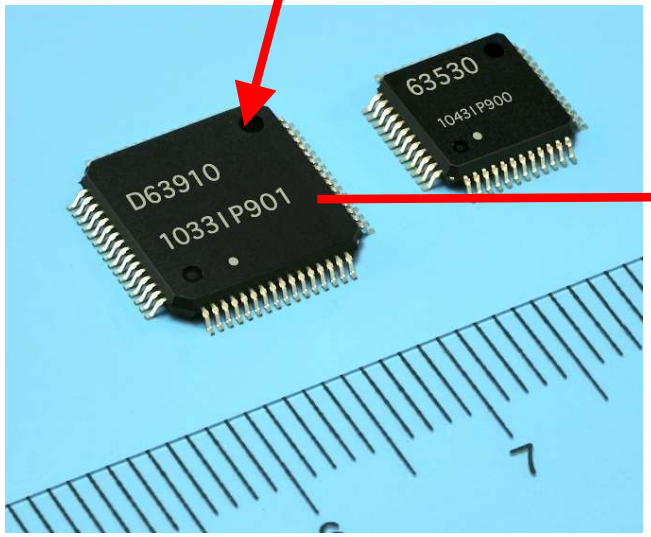




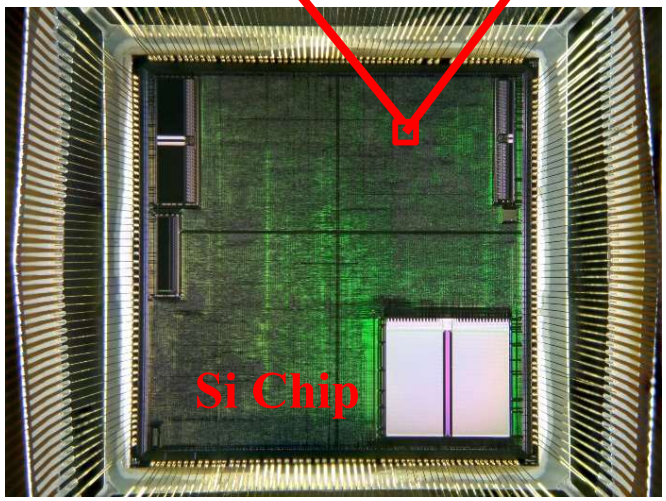
MOSFET







In the package・・・



Transistor

Type

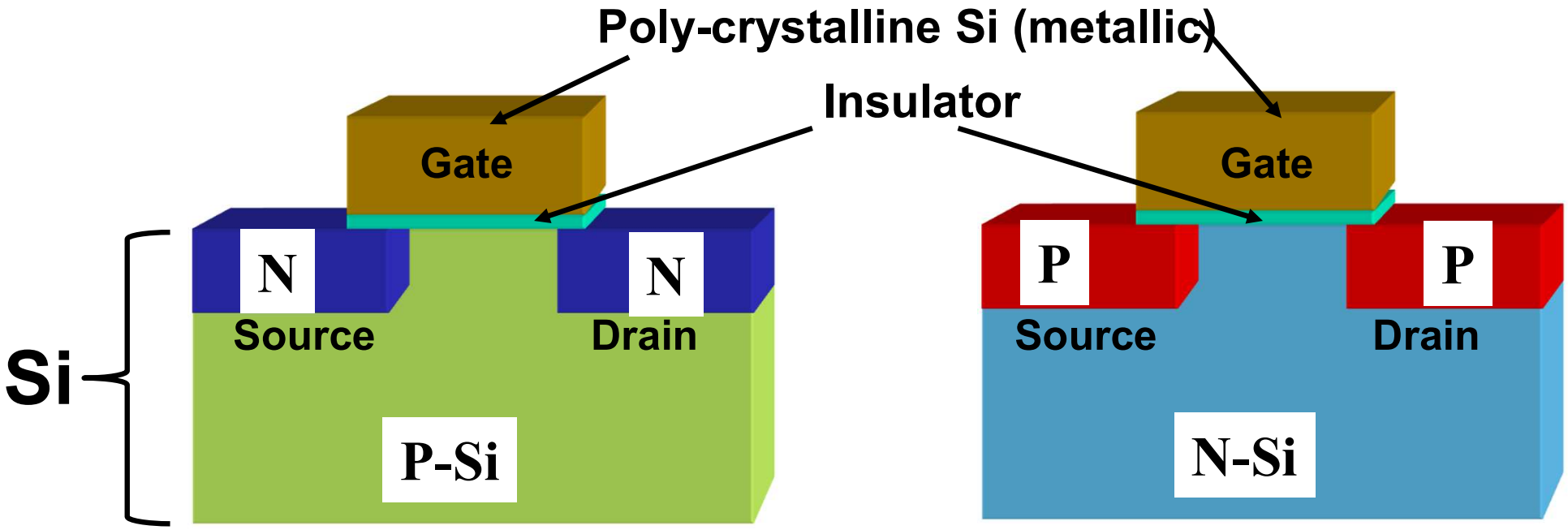
(1)Bipolar transistor (for large current)
(2)Unipolar transistor (for small signal)
Junction FET (Field effect transistor)
MOSFET

Main transistor in LSI

MOSFET
(Metal Oxide Semiconductor Field Effect Transistor)

MOSFET (cross section)

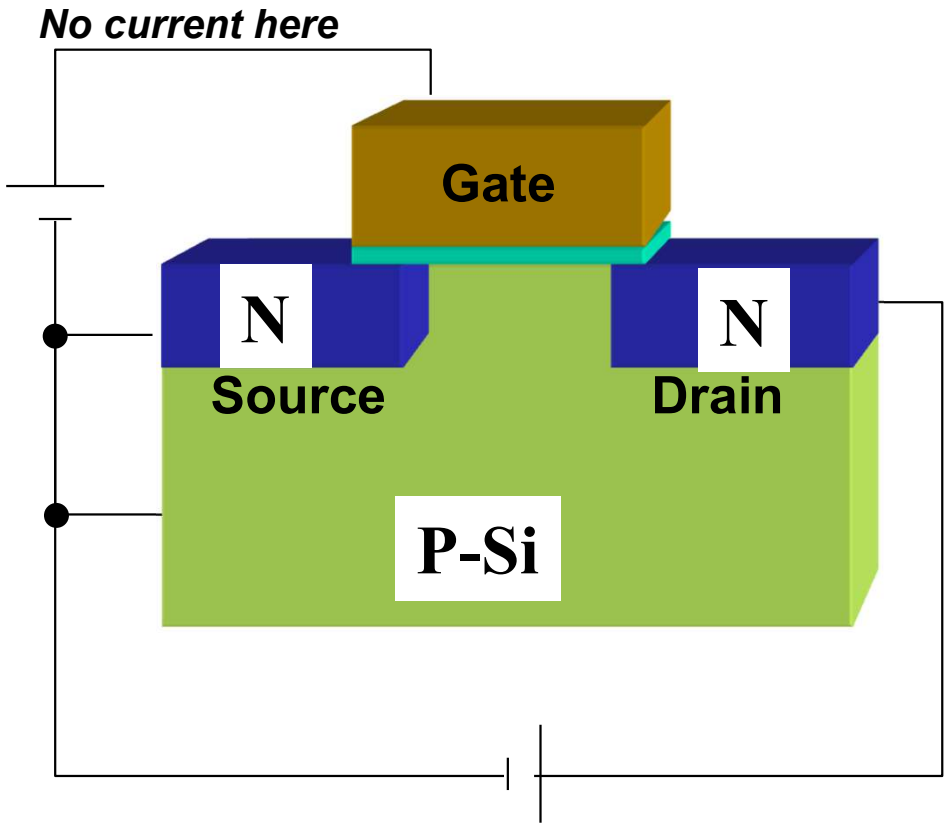
(Metal Oxide Semiconductor Field Effect Transistor)



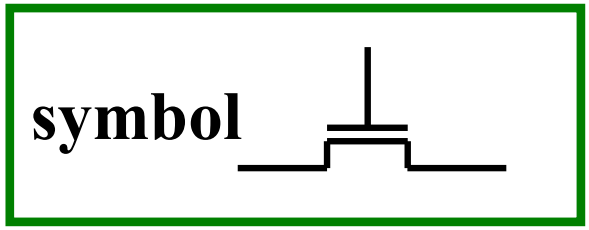
Stack structure, at middle, of Metal / Oxide) / Semiconductor → MOS

Semiconductor→material has an electrical conductivity value falling between that of a conductor, such as copper, gold etc. and an insulator, such as glass. Si, Ge, GaAs etc.

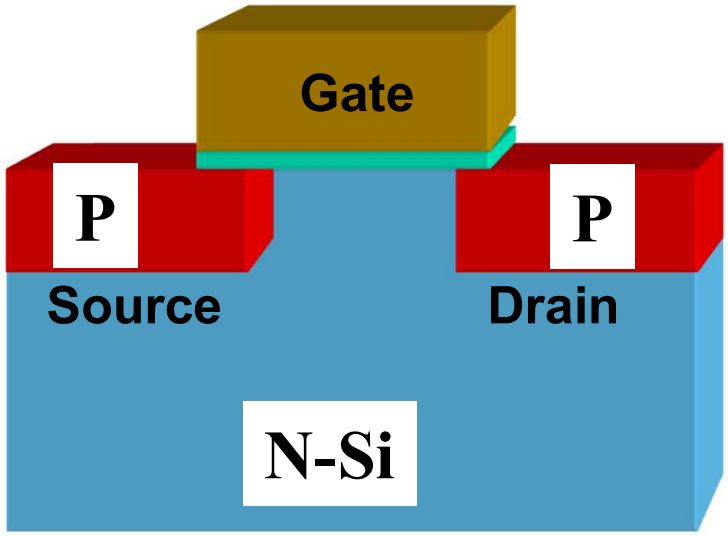
N-channel MOSFET (NFET)



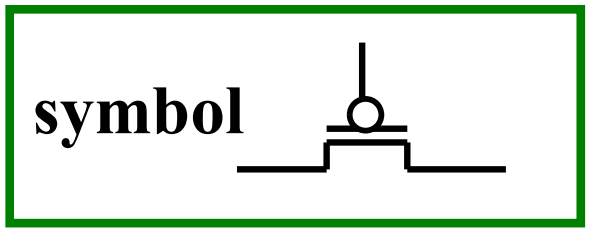
With positive bias to gate, resistance between source and drain becomes small.



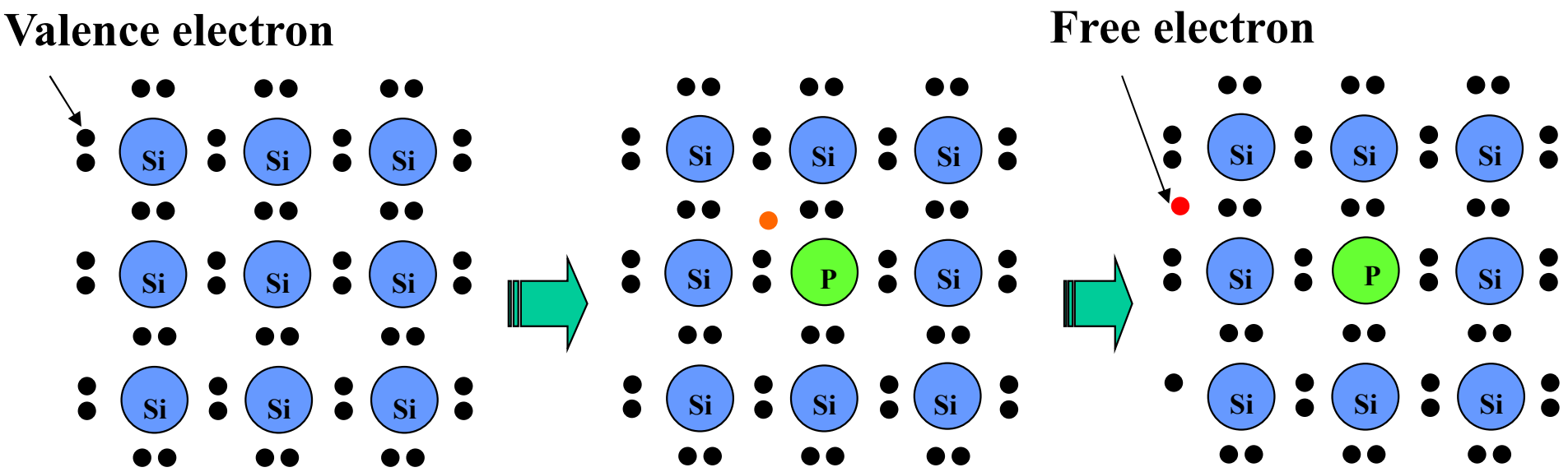
P-channel MOSFET (PFET)



With negative bias to gate, resistance between source and drain becomes small.



N-type Si



P(Phosphorus) incorporation: Ion implantation + thermal annealing

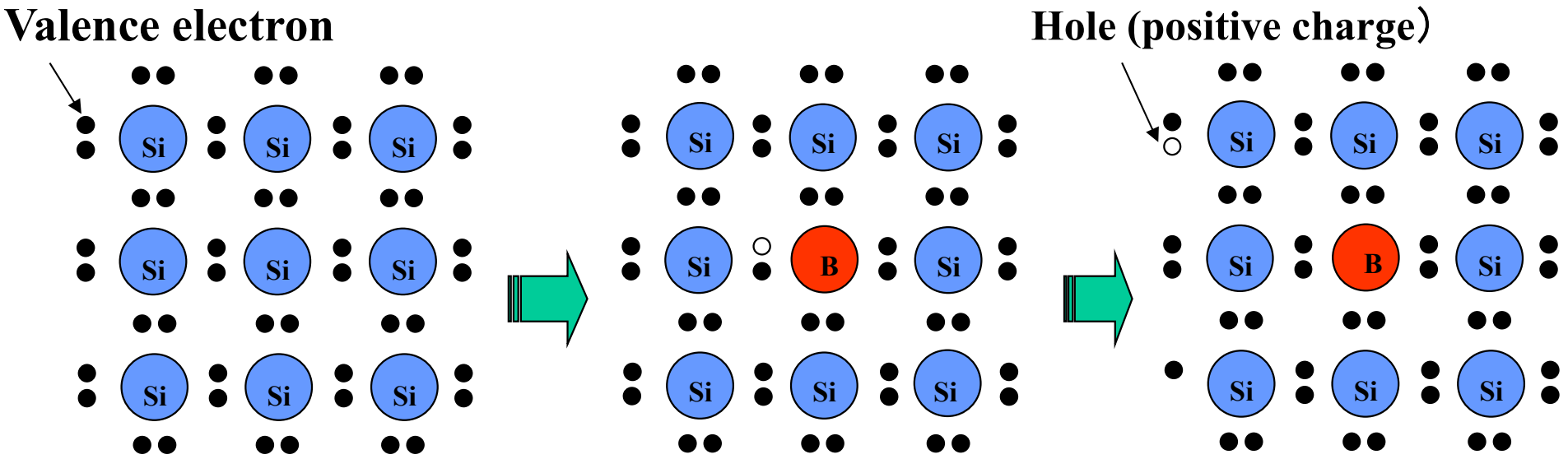
Very few free electrons in pure Si, $10^{10}/cm^3$

Number of valence electrons: $10^{23}/cm^3$

→ **High resistance**

⇒ **One free electron is generated with a P atom. P: donor**

P-type Si



B (Boron) incorporation: Ion implantation + thermal annealing

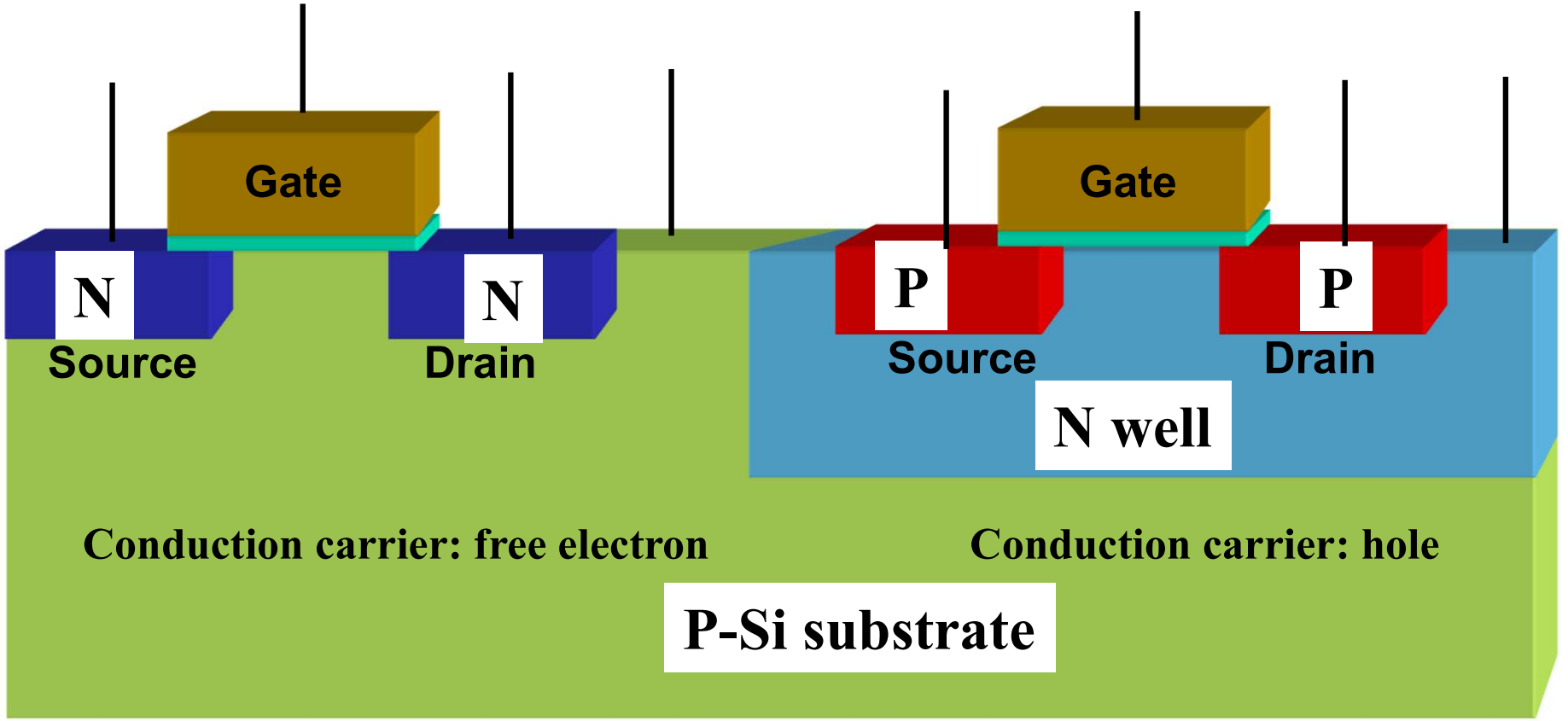
One valence vacancy (hole) is generated with a B atom.
Valence electrons can move via the vacancy.

B: acceptor

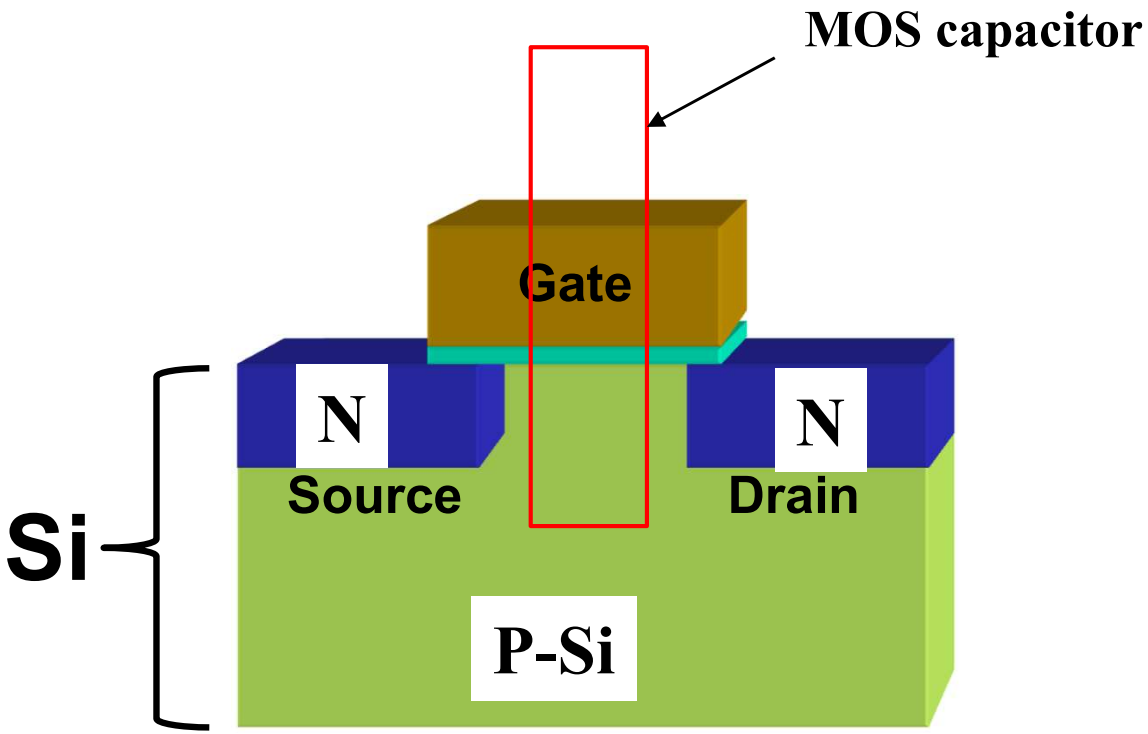
CMOS (Complimentary MOS)

N-channel MOSFET (NFET)

P-channel MOSFET (PFET)



We want you to have an experience of fabrication of MOS capacitor in this experiment.



Fabrication process of MOS capacitor

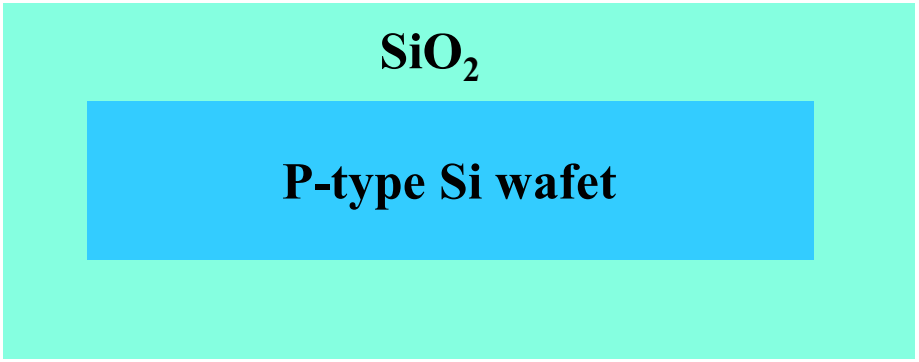
P-type Si wafet

Cleaning

Removal of
metal, organic contaminations
micro particles

Oxidation

O_2 ambient, at 1000C



Al deposition

Formation of Al thin film (40nm)

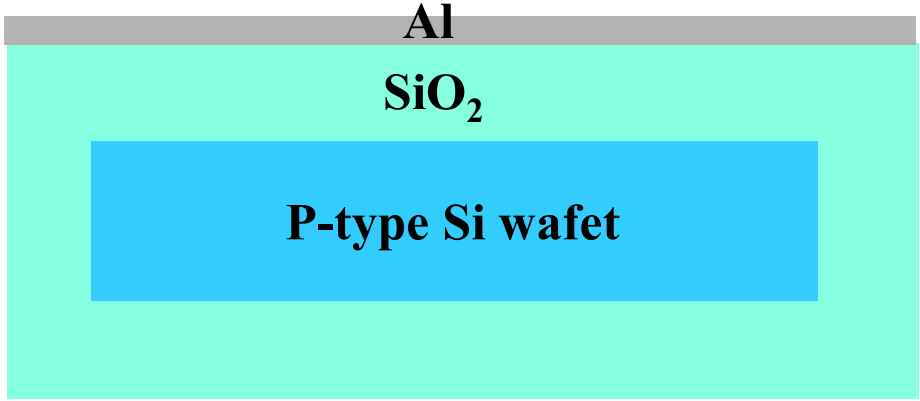
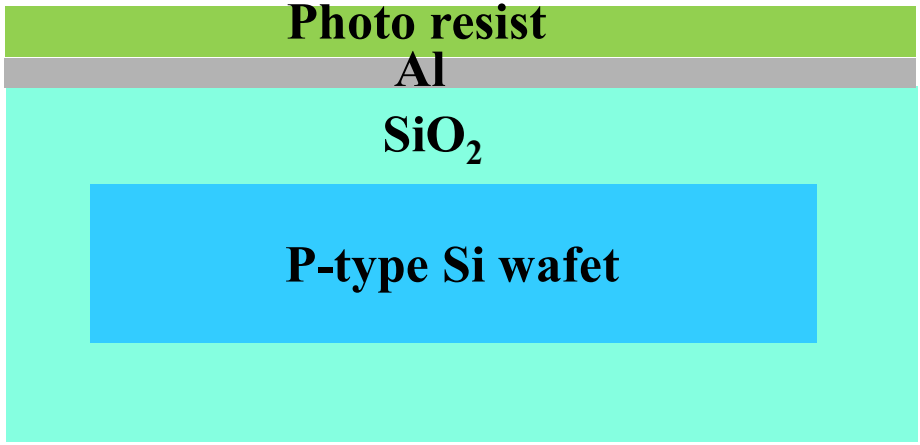
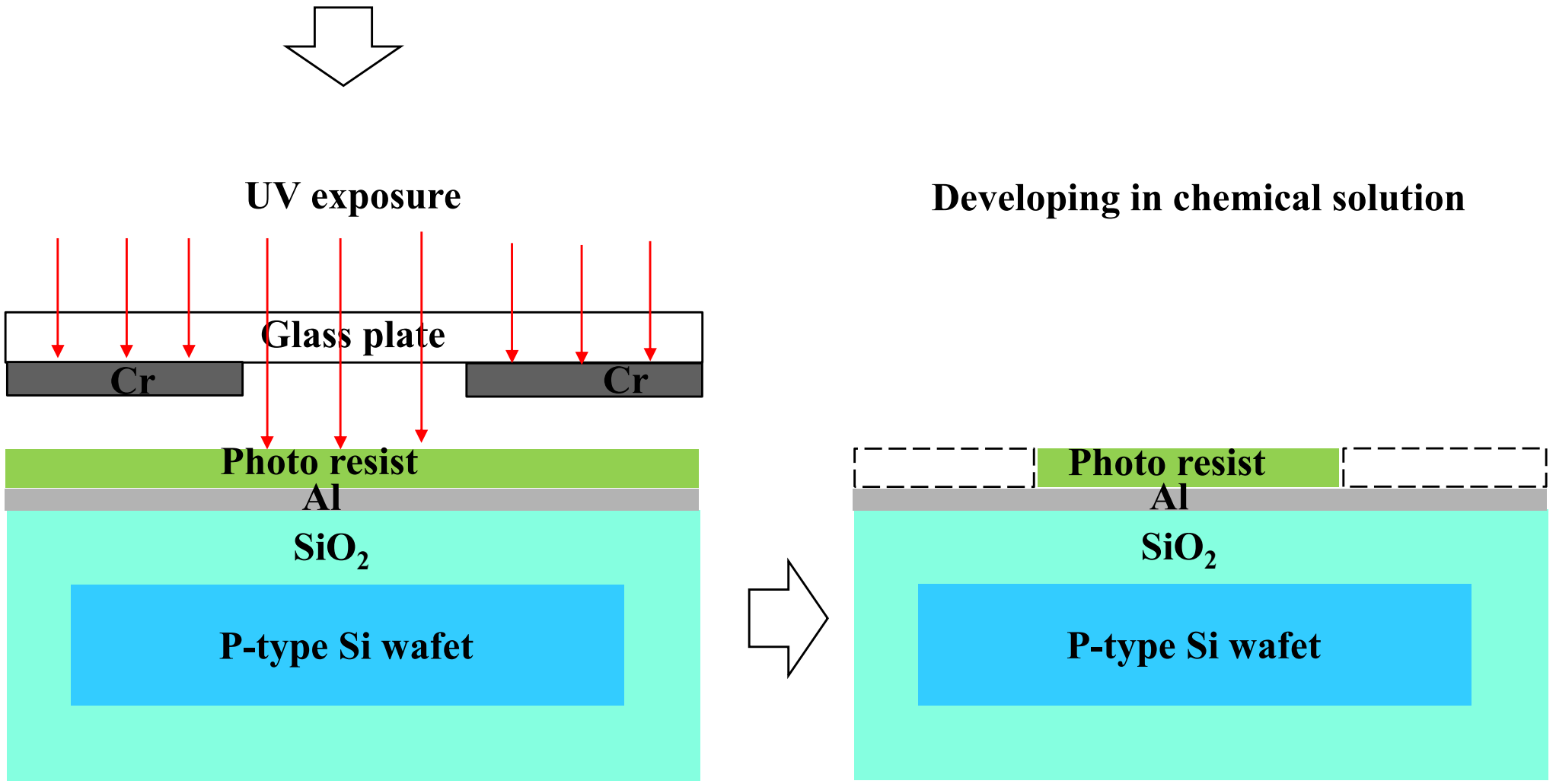
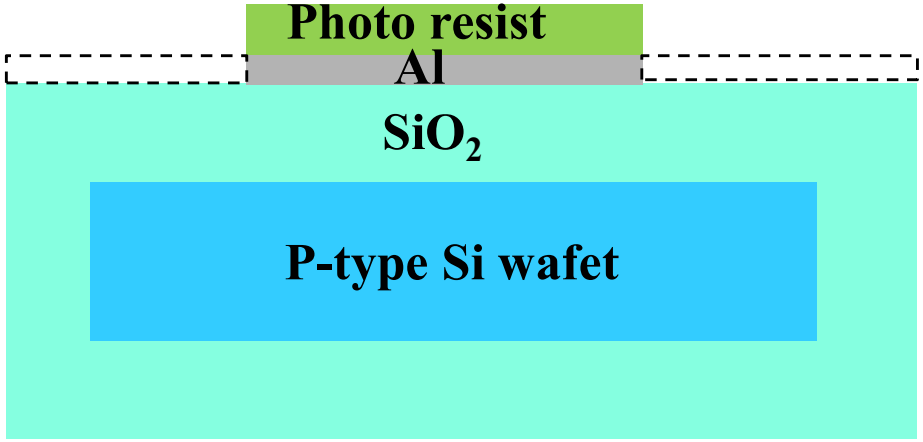


Photo resist (organic gel) coating

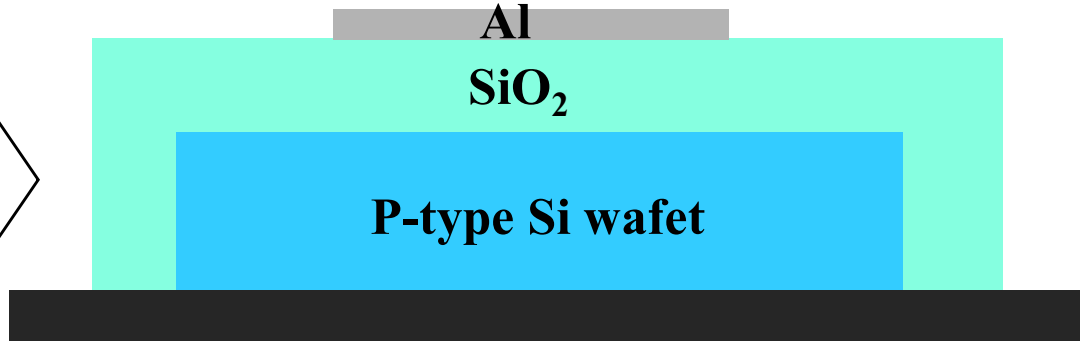



UV exposure
Glass plate
Cr
Cr
Photo resist
Al
SiO2
P-type Si wafet
Developing in chemical solution
Photo resist
Al
SiO2
P-type Si wafet
Only shaded area is removed in a certain solution

Al etching in acid



Removal of Phot resist and backside SiO_2
Metal plate attaching

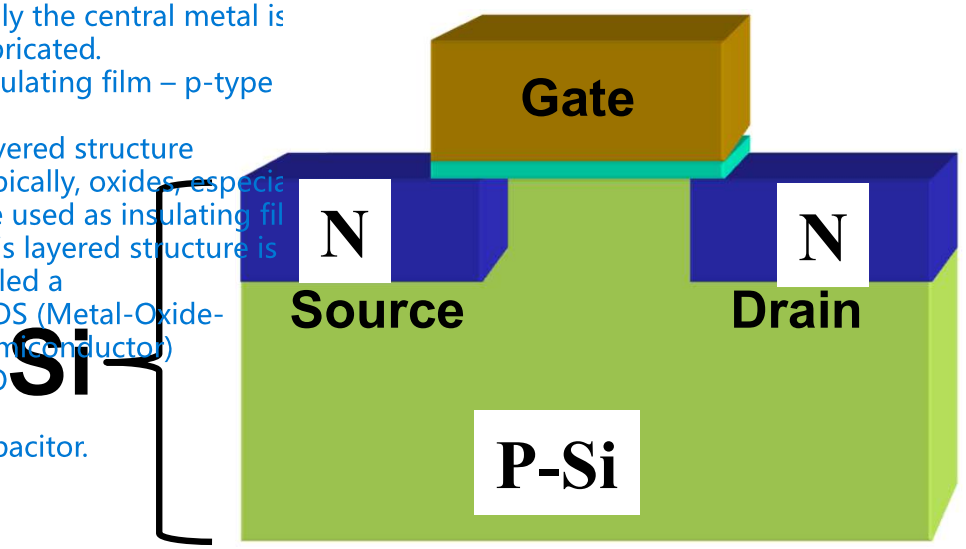


電子･量子主専攻　専攻実験
テーマD:
MOSと半導体/金属接合の作製と評価

担当 : 蓮沼　隆

テーマDではFET作製プロセスの一端を経験することを目的とする。

To create the source and drain on either side of the gate electrode, an ion implantation system is required. Unfortunately, University of Tsukuba does not have such equipment. Only the central metal is fabricated.
Insulating film – p-type Si
Layered structure
Typically, oxides, especially are used as insulating film. This layered structure is called a MOS (Metal-Oxide-Semiconductor) SiO_2 capacitor.



ゲート電極の両脇にあるソース、ドレインを作製するためには、イオンインプランテーション装置が必要。

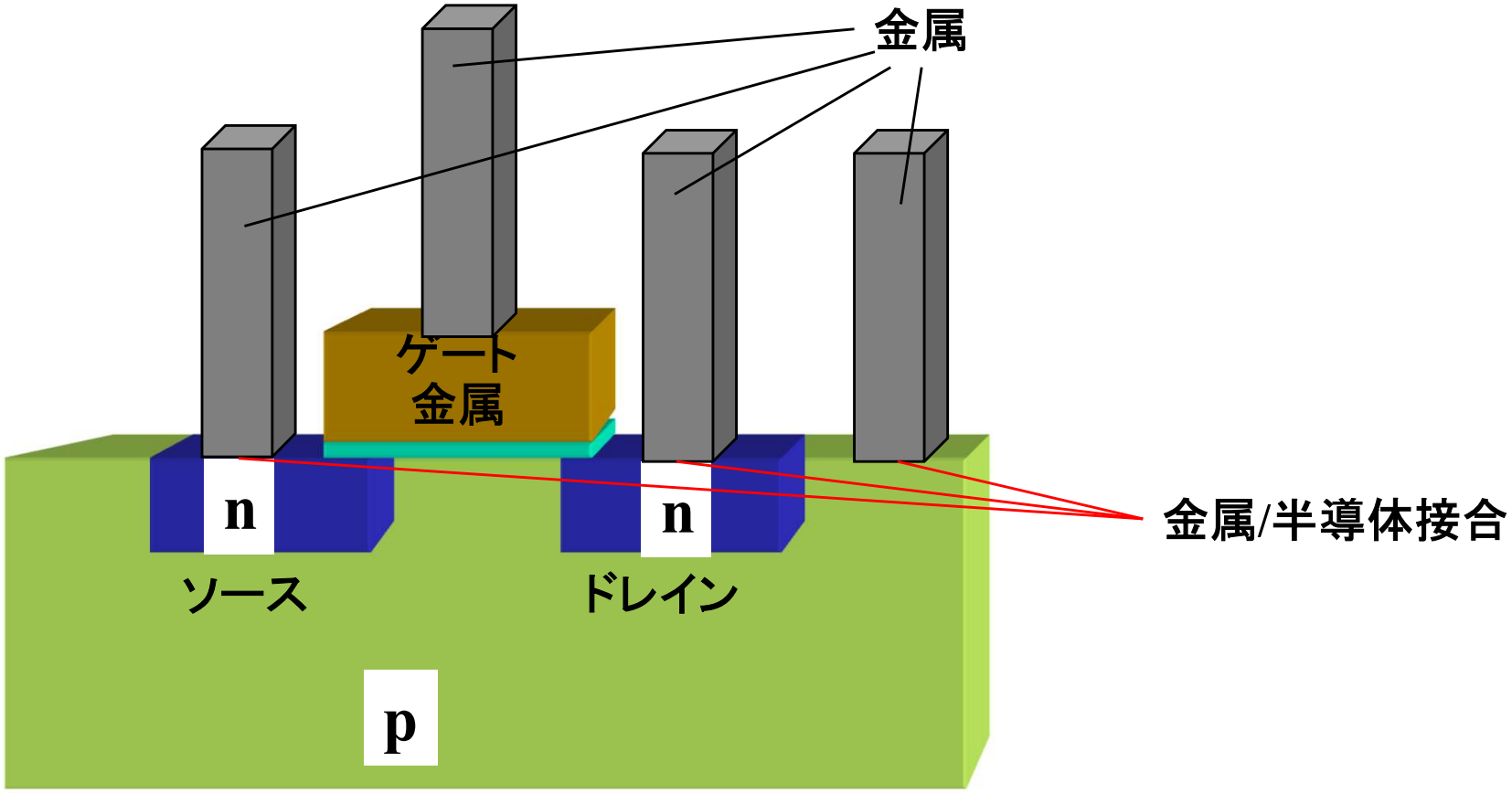
残念ながら筑波大には本装置がない。

中央の金属 - 絶縁膜 – p型Si 積層構造のみを作製

通常、絶縁膜として酸化物、特にSiO_2を用いることが多い。

この積層構造をMOS(Metal-Oxide-Semiconductor)キャパシタと呼ぶ

配線について



MOSFETは4端子素子であって、4か所の金属/個体接合がある。
通常、ゲートにも金属ではなく半導体を使うので、4か所の金属/半導体接合